

TX Squelch Architecture

Status: draft, for later integration into *Gizmo PMA Architecture Specification* after §8-4 (protocol squelch rows) and before §9 (TX Disparity Checker).

Status of the block. Proposed sequencer (**TxSquelchSeq**) plus two analog knobs (swing-mute DAC, per-lane V_{sq}).

Not yet in the behavioral model. CDNS pre-driver/serializer §3-1, voltage-mode driver §3-2, MRM §8, thermal-tuning loop **TBD_from_partner**. Numeric defaults that already appear in §8-4 (**Tsq_channel1**, **relink_squelch_tx_duration**) are reused; additional implementation targets are in the parameter table (§8-11).

Transmit squelch is a **whole-signal-chain behavior** spanning the pre-driver, driver, MRM, and MRM thermal-tuning (heater-lock) loop. It is **not** observe-only: a PMA-side sequencer commands the data-path source mux, the driver swing-mute DAC, and the squelch-bias trim. It is also **not** a second heater controller. The ring's thermal operating point remains owned by the thermal-tuning loop (one controller per node, §7-9 rule 1). Squelch's contract with that loop is: preserve the average-power error signal the servo already uses, freeze integrator ramp limits across the entry transient, and stay out of the heater-code path.

8-5 Purpose and why TX squelch exists

Squelch is the modulation-suppressed, **power-preserving** TX state mandated by OCI-MSA TXO-010 (Table 2-2, note 4): launched OMA ≤ -15 dBm per channel (**Tsq_channel1**, §8-4) while **average launched power remains constant**. It is not a quiet / idle convenience — it is a protocol signal.

Two distinct PMA uses share the same optical signature:

1. **Relink handshake.** The OCI-MSA deskew state machine uses squelch to command the far end to restart deskew: entering **Deskew_Data_Relink** squelches modulation on every WDM lane of the group for **relink_squelch_tx_duration** (60–75 ms, OCI-MSA Table 1-3 / §8-4), after which the transmitter proceeds **directly** into the 160-bit deskew training pattern. Handshake timing is a link-controller concern (the same split as the CDR's posture in §6-11); the PMA's job is to execute a glitch-free, constant-AOP mute and a phase-continuous exit into the commanded pattern.
2. **Invalid electrical input.** When the serializer-side stream is not mission data (static / idle / fault), the same mute holds average optical power so the TX and far-end RX ring heaters do not unlock. This is the use §8-4 currently names; it is the same plant, not a second mode.

The constant-average-power constraint exists because **both** ends of the link hold micro-ring resonances locked using average-power-derived error signals — TX modulator rings here, RX demux filter rings at the partner (**TBD_from_partner** on the far-end filter architecture). A squelch that disturbed average power would unlock heaters at both ends and turn a 75 ms handshake into a full optical re-acquisition. That is the same physics the disparity checker (§9) instruments in mission mode, taken to the limit of a fully static input.

Squelch is architecturally distinct from **PMD transmit disable** (OCI-MSA DJI-003): transmit disable removes optical power entirely via the ELS / laser path for safety and management, and does **not** preserve average power. Squelch shall never be implemented by ELS power reduction, heater park, or laser blanking. Gizmo does not yet specify the ELS disable path; the distinction is recorded here so a later management-plane section cannot reuse this sequencer.

8-6 Squelch principle for an MRM transmitter

In mission mode the MRM (§8-1) converts driver swing into optical amplitude by moving the ring resonance between two detuning states, producing levels P_1 and P_0 . For balanced data the launched average power is approximately

$(P_1 + P_0)/2$. Squelch must freeze the output at that average while removing the swing.

Two levers accomplish this:

1. **Electrical modulation mute** in the pre-driver / driver (swing suppression). Primary mute is commanded at the pre-driver (swing-DAC to minimum, or data forced to a legal static hold code — §3-1 "Serializer legal static states"). The driver contributes the balance of the mute. The composite electrical target is ≥ 20 dB of swing suppression, which, relative to the §8-2 maximum operating OMA of -1 dBm, yields ≥ 14 dB of optical OMA suppression and lands on $T_{sq_channel} \leq -15$ dBm with margin.
2. **Calibrated static bias** V_{sq} applied through the driver's DC path, chosen so the ring's static transmission equals the time-averaged transmission of the modulated state.

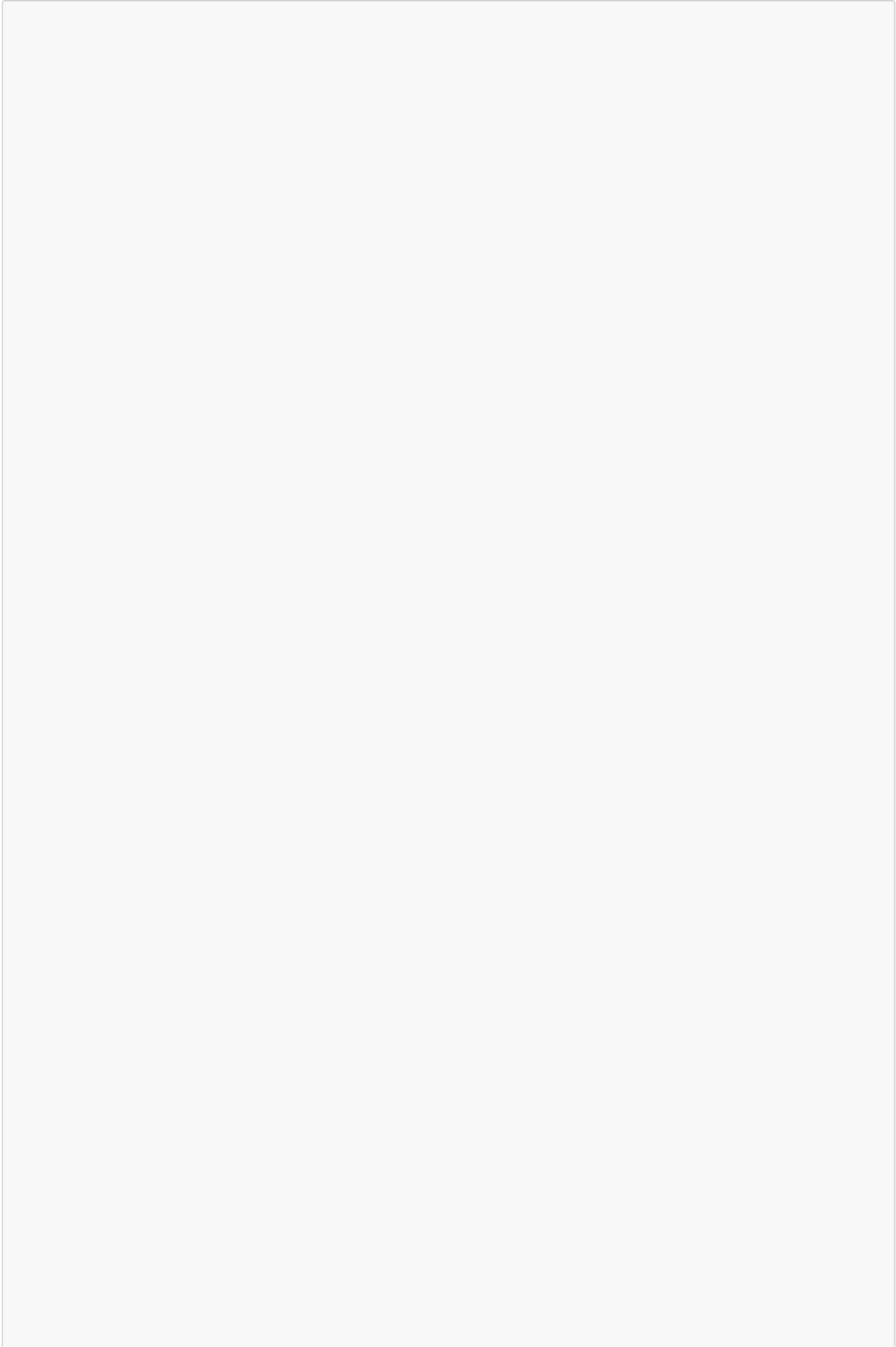
Because the ring transfer function is strongly nonlinear in both voltage and wavelength, V_{sq} is **not** the electrical mid-swing point and cannot be derived analytically with sufficient accuracy. It is a **per-channel calibrated quantity**, trimmed against the TX monitor photodiode (PIC monitor path $TBD_from_partner$ / TBD_analog_design). It is a distinct operating point from the mission DC reverse-bias V_{bias} (-1.5 V to -2.0 V, §8-3).

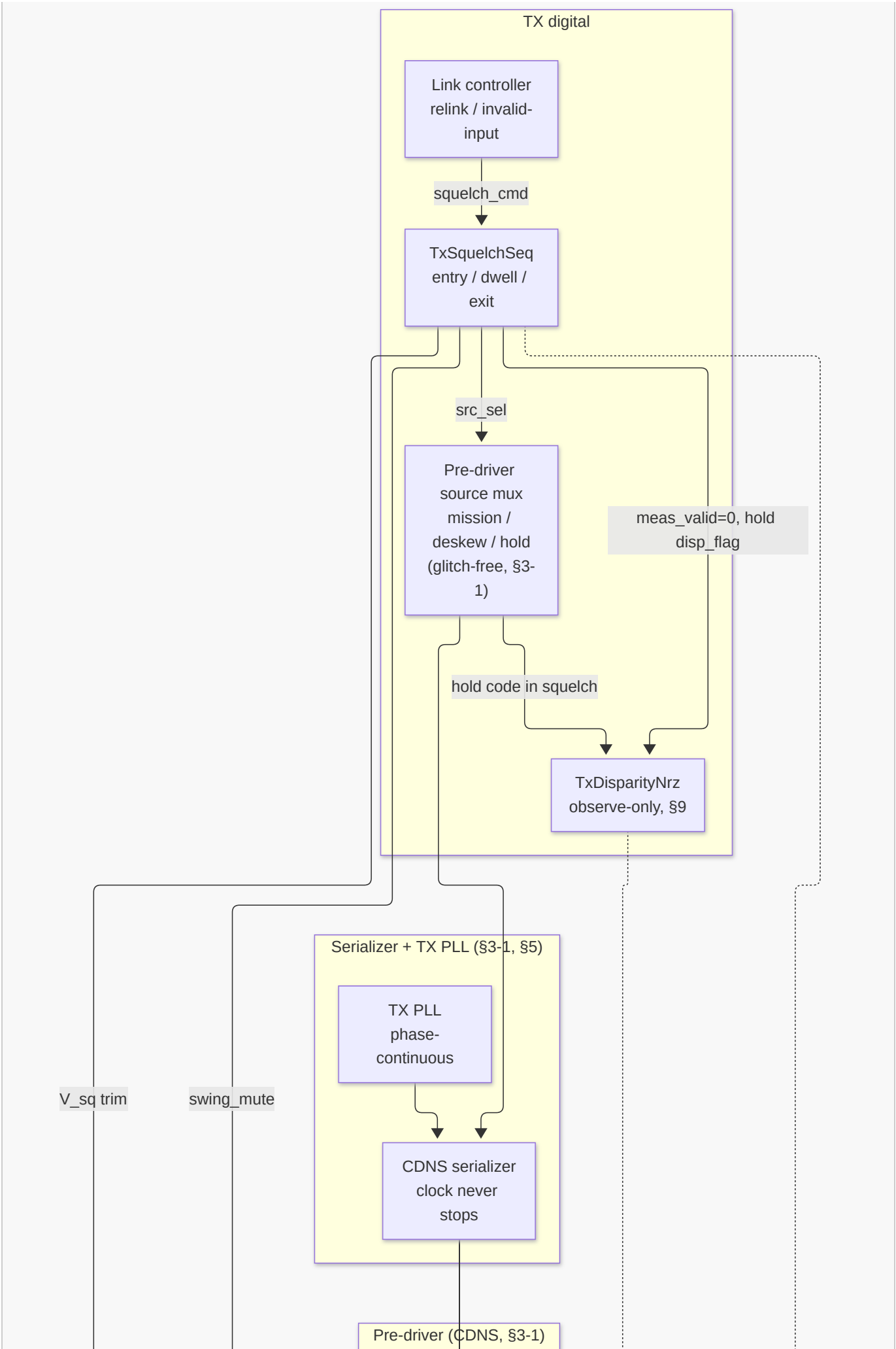
Two second-order effects must be absorbed by the heater servo, not by opening it:

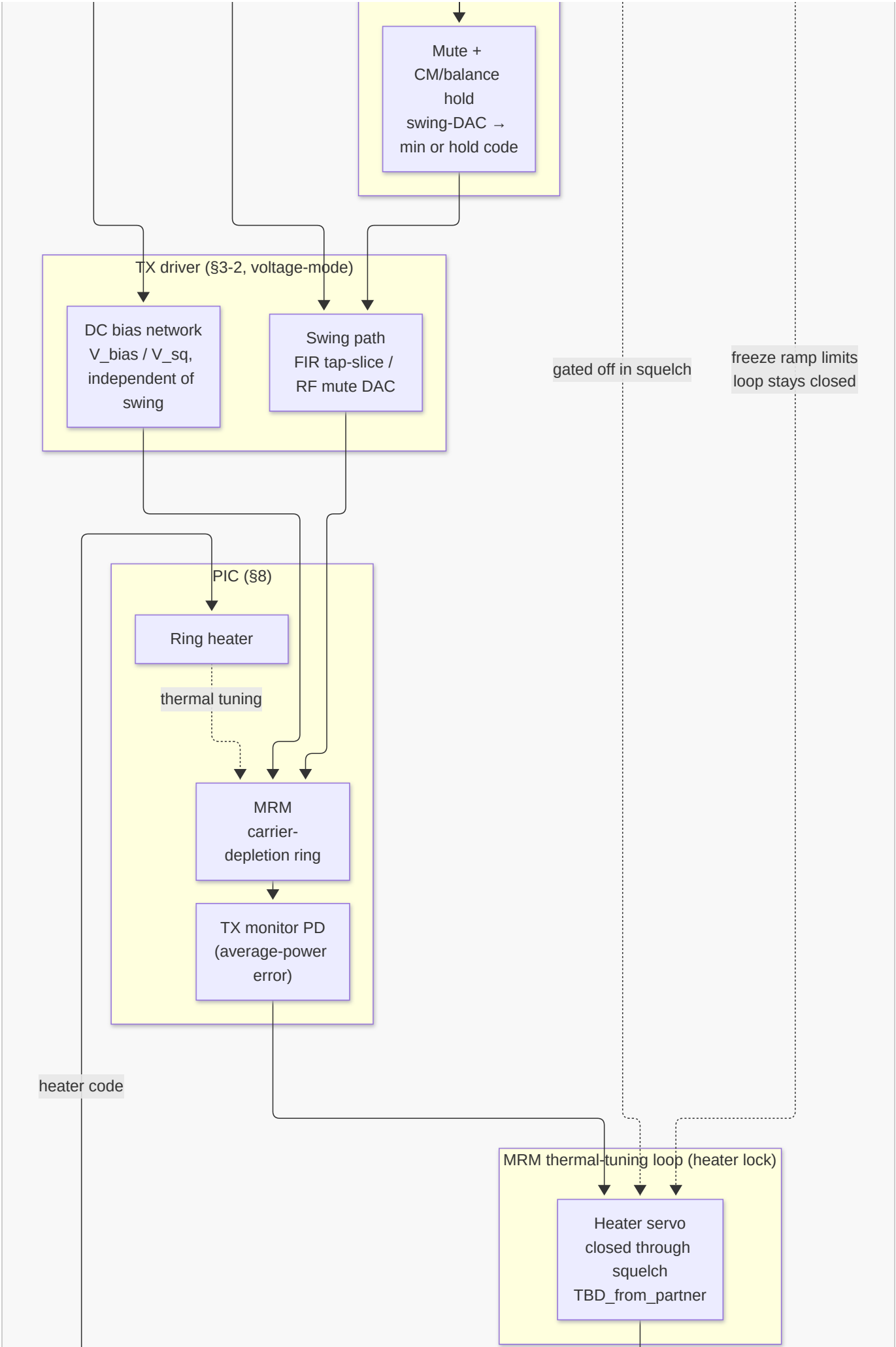
- Removing RF modulation changes the average intracavity optical energy and the carrier-induced self-heating of the junction, producing a small resonance step at squelch entry and exit. This is the same data-dependent self-heating mechanism §9-1 names, now a step rather than a density drift.
- The heater servo's error signal is the monitor-photodiode **average power**, which squelch preserves by construction — this is why TXO-010 forbids average-power squelch. The loop, including its lock dither, runs uninterrupted through entry, dwell, and exit.

DAC terminology. As in §7, "DAC" here means a **digital code controlling an analog setting**, not necessarily a voltage DAC. The swing-mute code, the V_{sq} trim, and the heater code are three different DACs on three different nodes. Squelch may write the first two; it shall not write the heater code.

8-7 Block diagram and block-level responsibilities







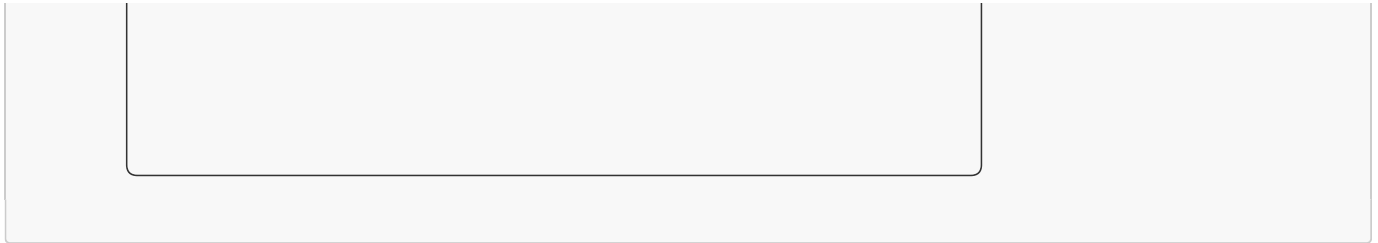


Figure 8-2: TX squelch spans pre-driver source select and mute, driver swing-mute (independent of the DC bias network), per-channel V_{sq} on the MRM, and the heater servo. The TX PLL and serializer keep running. The disparity checker is gated, not used as a substitute error signal. The heater code remains owned by the thermal-tuning loop.

Pre-driver (CDNS, §3-1).

- **Data-path source select:** a glitch-free three-way multiplexer (mission data / deskew training pattern / squelch hold code), switched synchronously to the serializer clock. The TX PLL (§5) and serializer **never stop**; only the data content changes. That is what guarantees the phase-continuity the deskew protocol requires of pattern transitions, and what the far-end CDR's signal-valid / warm re-acquire path (§6-11) depends on.
- **Mute control:** primary swing suppression is commanded here — swing-DAC to minimum, or data forced to the hold code — sequenced with the driver mute to reach the composite ≥ 20 dB electrical suppression target. The legal static (non-toggling) serializer states this mute uses are the §3-1 row already tagged [TBD_from_partner](#).
- **Common-mode and balance hold:** pre-driver output common mode and duty-cycle balance are held through mute entry/exit so no common-mode step propagates to the modulator and perturbs AOP beyond the ± 0.5 dB window. This is the same CM/DCD allocation already levied on CDNS in §3-1, now with an explicit squelch-entry constraint.

Driver (§3-2).

- **Independent swing and bias paths:** the output stage shall separate the RF swing (the analog-FIR tap-slice / swing-mute DAC) from the DC bias network applied across the MRM junction, so that muting the swing leaves the modulator DC operating point untouched. Gizmo's driver is **voltage-mode** (§3-2). A topology in which swing and bias share a control (for example supply-collapse mute) is prohibited, because it moves the ring operating point and violates constant AOP. Voltage-mode mute is therefore a swing-path disable or FIR-slice collapse to a DC hold, **not** a rail collapse ([TBD_analog_design](#)).
- **Mute depth and settling:** driver mute contributes the balance of the ≥ 20 dB electrical suppression, with entry/exit settling ≤ 1 ms so the driver transient is negligible against the 60 ms minimum squelch dwell.
- **Residual-swing floor:** output-stage feedthrough (clock, data-path leakage, FIR-slice residual) at the launched-OMA level must remain below -15 dBm optical. Isolation/feedthrough is a driver specification ([TBD_analog_design](#)).

MRM (§8).

- **Calibrated squelch bias V_{sq} per channel,** applied through the driver DC path, placing static transmission at the mission-mode average. Factory-set and periodically re-trimmed in service using the monitor photodiode, because V_{sq} drifts with ring detuning, temperature, and aging.
- **Operating-regime containment:** V_{sq} shall keep the junction in its normal depletion regime (§8-1 / §8-3) so carrier-plasma index/absorption changes between squelched and modulated states stay small and the resulting resonance step remains inside the heater servo's linear capture range.

MRM controller — heater servo (thermal-tuning loop, [TBD_from_partner](#)).

- **Continuous closed-loop lock:** the servo's error signal is average power at the monitor photodiode, which squelch preserves by construction. The loop, including its lock dither, runs through entry, dwell, and exit.

Squelch does not substitute a disparity-checker density, a frozen heater code, or an open-loop park.

- **Dither budget in squelch:** heater dither converts to residual OMA at the output. Dither amplitude during squelch shall be bounded so dither-induced OMA does not consume `Tsq_channel`. If mission-mode dither is too large, a reduced-dither squelch profile shall be defined — with the corresponding lock-slope penalty analyzed (`TBD_from_sim_sweep`).
- **Thermal-step absorption:** the servo shall settle the squelch entry/exit resonance step to within the mission detuning budget in ≤ 10 ms (target), inside the 60 ms minimum dwell, with zero unlock events over the qualification soak.
- **Far-end dependency:** the same constant-AOP property keeps the remote RX demux-filter servo locked. Any AOP droop during our squelch is a direct attack on the partner's lock margin, which is why AOP is budgeted at ± 0.5 dB rather than at a 3 dB LOS margin. Relative to the §8-2 `Pavg` floor of -8.5 dBm, a ± 0.5 dB squelch excursion does not approach a typical far-end LOS assert (OCI-MSA RXO-006, -14 dBm AOP).

Lane atomicity. All WDM lanes of a PMA group shall enter and exit squelch together, with lane-to-lane entry/exit alignment such that the far-end deskew engine observes a consistent group state. Per-lane squelch is reserved for test modes only. Lane count is `TBD_from_partner` (§3-1 serializer lane count, §8-2 `Pavg_total`).

8-8 Algorithm and state machine

The sequencer `TxSquelchSeq` is commanded by the link controller. It owns the pre-driver `src_sel`, the driver swing-mute DAC, and the V_{sq} trim. It does not own the heater code, the TX PLL, or the serializer clock.

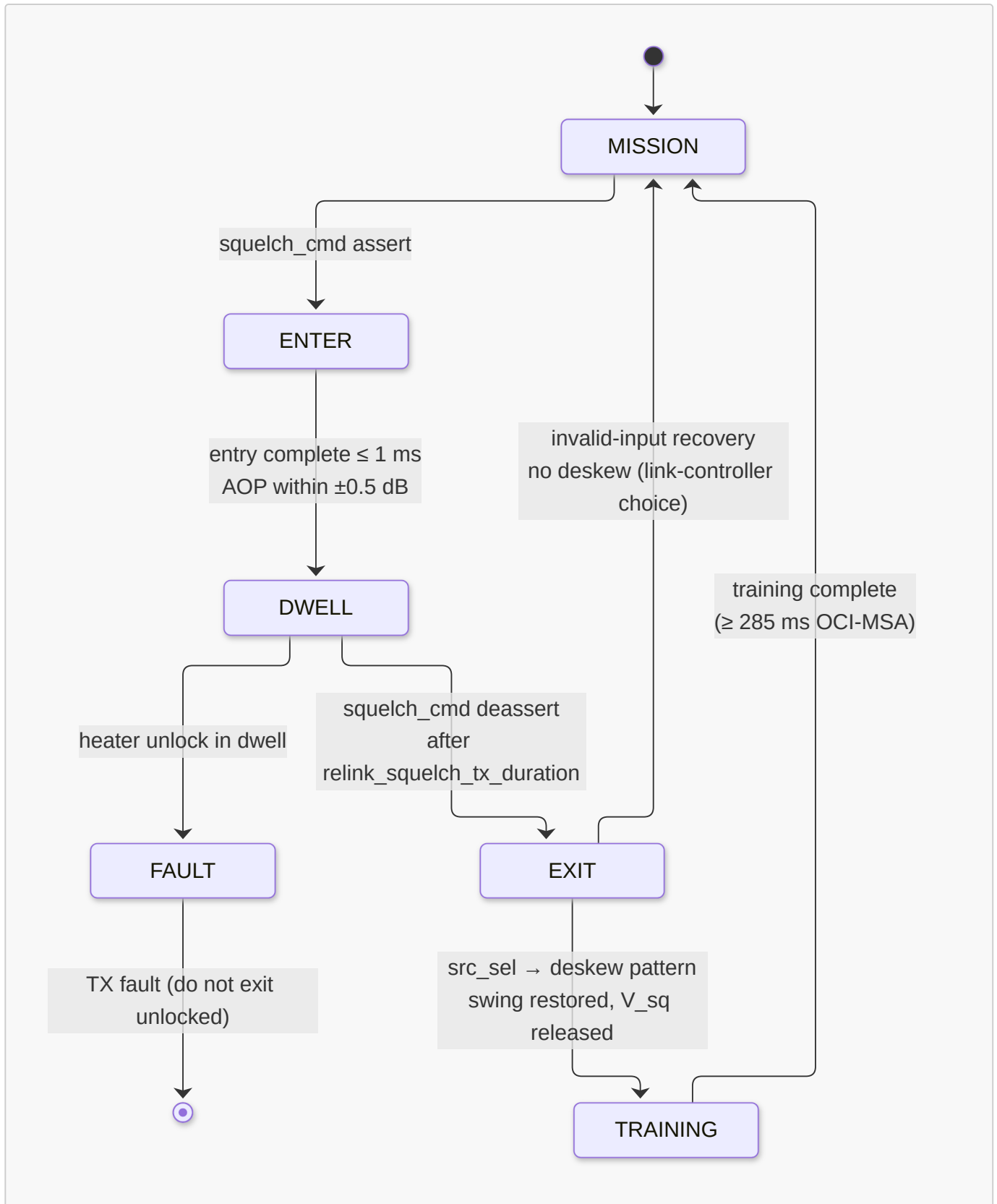


Figure 8-3: TX squelch sequencer. Entry and exit are analog/digital sequences on a running serializer; DWELL is the protocol interval. Unlock during dwell escalates to TX fault rather than returning an unlocked ring to mission modulation.

Algorithm (TxSquelchSeq).

```

# Owns: src_sel, swing_mute DAC, V_sq trim
# Does not own: heater code (thermal-tuning loop, one controller per node, §7-9)

```

```

def enter(t0):
    heater.freeze_integrator_ramp_limits() # transient guard; loop stays CLOSED
    disparity.meas_valid = 0 # §9-4: not mission data
    hold(disparity.disp_flag)
    src_sel = HOLD # glitch-free; serializer + PLL
    keep running
    ramp(swing_mute, MUTE) # independent of V_bias / V_sq
    apply(V_sq) # per-channel calibrated static
    bias
    # complete within t_entry ≤ 1 ms; launched AOP within ±0.5 dB of pre-squelch
    state = DWELL

def dwell():
    # servo re-settles the RF-removal thermal step (target ≤ 10 ms)
    # dither continues at the squelch profile (bounded vs Tsq_channel)
    # residual OMA verified ≤ -15 dBm; all WDM lanes of the group stay muted
    if heater.unlock:
        escalate_tx_fault() # do not exit squelch with an
    unlocked ring
        state = FAULT
        return
    # far end detects loss of modulation within t_loselock ≤ 50 ms (OCI-MSA)

def exit_to_training():
    src_sel = DESKEW_TRAINING # phase-continuous; 160-bit OCI-
    MSA pattern
    ramp(swing_mute, MISSION)
    release(V_sq) # back to mission V_bias
    disparity.acc = 0; disparity.persist = 0 # §9-4: first post-squelch
    snapshot is clean
    disparity.meas_valid = 1 # once the training/mission stream
    is live
    heater.release_integrator_ramp_limits()
    state = TRAINING # ≥ 285 ms training (OCI-MSA),
    then MISSION

```

Abort / fault. If the servo reports unlock during dwell, the channel escalates to TX fault rather than exiting squelch with an unlocked ring. The PMA does not attempt to re-acquire the heater; that would put a second controller on the thermal node.

Mapping to the common architecture. This is a sequencer plus two DACs (swing-mute, V_{sq}), not a §7-1 vote → scale → accumulate loop. The heater servo is the loop; squelch is a plant change that loop must survive. The disparity checker remains observe-only and is **gated** for the duration (§9-4).

8-9 Timing

Interval	Symbol	Target	What must be true
Entry sequence	t_{entry}	≤ 1 ms	Hold-code switch, swing mute, V_{sq} apply, CM/balance hold; AOP excursion within ±0.5 dB
Heater thermal-step settle	$t_{\text{th},sq}$	≤ 10 ms (target)	Residual detuning back inside the mission detuning budget; zero unlock

Interval	Symbol	Target	What must be true
Relink squelch dwell	<code>relink_squelch_tx_duration</code>	60 ms to 75 ms	OCI-MSA Table 1-3 / §8-4. Modulation stays suppressed for this entire interval. Entry and thermal settle are inside this window, not in addition to it
Far-end loss-of-modulation detect	<code>t_loselock</code>	≤ 50 ms	Bounds allowable residual OMA ripple during dwell; far-end CDR takes the §6-11 signal-valid gate
Exit sequence	t_{exit}	≤ 1 ms (same mute-settling target)	Phase-continuous source switch to the 160-bit deskew training pattern, swing restore, V_{sq} release; immediately compliant with the ≥ 285 ms training-pattern transmission requirement
Deskew training after exit	—	≥ 285 ms (OCI-MSA)	Link-controller / PCS concern; PMA must not insert an extra gap between squelch exit and the training pattern



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Figure 8-4: Relink squelch timing. The 60–75 ms figure is the **dwell** at squelched OMA, not the time allowed to slew into or out of squelch.

`relink_squelch_tx_duration` is the **dwell** of the handshake: 60 ms minimum so the far end can detect mute inside a 50 ms `t_loselock` and still see a 10 ms settle margin; 75 ms maximum before training must start. Enter/exit settling is a separate ≤ 1 ms budget. (The §8-4 notes cell currently reads this interval as "duration to enter/exit" and says "MRR"; both should be corrected on merge — dwell, and **MRM**.)

The 60–75 ms window is comfortable against the thermal plant assumed in §9-6 (heater-control settling ms-class, `τ_th` μs-class): a 10 ms thermal-step settle is well inside a 60 ms minimum dwell. If partner data move `τ_th` or the heater-loop bandwidth enough to threaten that 10 ms target, the dwell cannot be shortened — it is an OCI-MSA protocol number — the servo and V_{sq} trim have to close instead (`TBD_from_partner`).

8-10 Interfaces

Link controller. Assert/deassert `squelch_cmd`, choose the exit destination (deskew training vs. return to mission), and own `relink_squelch_tx_duration` within the 60–75 ms window. The PMA does not time the handshake itself — same split as §6-11.

CDNS serializer / pre-driver (§3-1, §5). Legal static hold code; glitch-free `src_sel`; TX PLL and serializer remain in lock and in phase through the whole sequence. Phase-continuity on hold → training is what keeps the far-end CDR from treating exit as a cold re-acquisition.

Driver swing-mute and bias DACs (§3-2, §8-3). Two knobs, two nodes. Squelch writes swing-mute and V_{sq} ; it does not write FIR tap weights as a substitute for mute (tap codes stay at their mission values so exit is hitless — the existing glitchless-FIR rule, §3-2), and it does not write the heater code.

TX disparity checker (§9-4). While `squelch_cmd` is asserted the serializer input is not mission data. A disparity measured on a static hold code would look like a sustained `|dens_meas| = 1` event and must not reach the thermal-tuning loop, which is at that moment relying on the constant-AOP squelch state to hold heater lock. Follow the CDR's signal-valid discipline (§6-11): `meas_valid` forced low, `disp_flag` held; on exit, window accumulator and persistence counters cleared so the first post-squelch snapshot is not a partial window. This section is the definition of the "TX-side squelch/invalid condition" that §9-4 already names.

Thermal-tuning loop. Closed throughout. At entry, freeze integrator **ramp limits** (slew/anti-windup guard) so the 1 ms electrical transient is not integrated as a huge error; do not freeze the integrator state and do not open the loop. Optionally switch to a reduced-dither squelch profile. Unlock in dwell → TX fault, not a PMA takeover of the heater DAC.

Far-end RX (informative to this PMA). Our squelch is the far-end's invalid-signal condition: that CDR holds `pi_code` / `state_p` / `state_f` (§6-11) rather than drifting on a muted input. Far-end demux-filter heaters stay locked because AOP is constant. Far-end adaptation loops freeze per §7-11 signal-invalid hold. None of that is implemented here; it is why our AOP and residual-OMA budgets exist.

Lane group. `squelch_cmd` is lane-atomic across the WDM group. Per-lane override is a test-mode hook only.

8-11 Parameter table

Placeholder / symbol	Model/RTL name	Default	Meaning
<code>Tsq_channel</code>	<code>tsq_channel</code>	≤ -15 dBm OMA per channel	Squelched launched OMA (OCI-MSA TXO-010 / §8-4)
<code>relink_squelch_tx_duration</code>	<code>relink_squelch_tx_duration</code>	60 ms to 75 ms	OCI-MSA Table 1-3 dwell at squelched OMA, not the enter/exit slew time
$\Delta P_{avg,sq}$	<code>aop_squelch_tol</code>	± 0.5 dB vs. pre-squelch <code>Pavg</code>	AOP constancy across entry/dwell/exit. Compatible with §8-2 <code>Pavg</code> (−8.5 dBm to 0 dBm)
$A_{mute,e}$	<code>swing_mute_db</code>	≥ 20 dB electrical (design target)	Pre-driver + driver composite swing suppression. Optical consequence: ≥ 14 dB OMA suppression vs. §8-2 max OMA −1 dBm → <code>Tsq_channel</code>
t_{entry}, t_{exit}	<code>t_squelch_settle</code>	≤ 1 ms	Mute/bias sequencing and CM/balance settle
$t_{th,sq}$	<code>t_heater_squelch_settle</code>	≤ 10 ms (target)	Heater-servo absorption of the RF-

Placeholder / symbol	Model/RTL name	Default	Meaning
			removal/restoration thermal step
<code>t_loselock</code>	—	≤ 50 ms (far end, OCI-MSA)	Far-end loss-of- modulation detect; bounds residual OMA ripple in dwell. Not a local timer
V_{sq}	<code>v_sq</code> (per lane)	calibrated, ≠ mid- swing; ≠ mission V_{bias}	Static MRM bias whose transmission equals modulated-state average. Factory + in- service trim vs. TX monitor PD (<code>TBD_from_partner</code>)
<code>src_sel</code>	<code>src_sel</code>	{MISSION, DESKEW, HOLD}	Glitch-free three-way source mux. HOLD uses the §3-1 legal static serializer state (<code>TBD_from_partner</code>)
<code>swing_mute</code>	<code>swing_mute</code>	{MISSION, MUTE}	Driver RF-swing mute DAC, independent of the DC bias network (<code>TBD_analog_design</code> for voltage-mode realization)
—	<code>squelch_cmd</code>	level from link controller	Lane-atomic command; per-lane override is test- only
—	<code>squelch_dither_profile</code>	mission dither, or reduced	Heater dither during dwell; must not consume <code>Tsq_channel</code> (<code>TBD_from_sim_sweep</code>)
—	<code>heater_ramp_limit_freeze</code>	asserted in ENTER/EXIT	Transient guard on the servo integrator; loop remains closed

Dead-band / hysteresis (TX squelch). Squelch is a commanded sequencer, not a bang-bang loop, so it carries **no vote dead-band** of the §7-12 kind.

What *does* sit on this path:

- **AOP compliance window ±0.5 dB** — a plant-excursion limit during entry/dwell/exit, not a loop dead-band. The heater servo's own (unspecified) lock detector and dither live inside that window (`TBD_from_partner`).
- **Heater dither vs. `Tsq_channel`** — dither is residual OMA. If mission dither would violate −15 dBm, switch profiles rather than opening the loop.

- **Disparity-checker hysteresis is not used during squelch.** `disp_flag` is held and `meas_valid` is low (§9-4, §7-12). Re-enable only after the post-exit accumulator clear.
- **Integrator ramp-limit freeze** at entry/exit is a de-glitch strobe on the thermal loop (same idea as the AGC/CTLE de-glitch strobes in §7-9), not a second controller.

Nesting / disturbance ladder. Squelch is TX-side and ms-class. To the RX ladder it is an invalid-signal event: the far-end CDR holds (§6-11) and every continuous RX adaptation loop freezes (§7-11). Locally, the only loop that **keeps running** is the heater servo, which is already the slowest plant in the document (μs τ_{th} , ms settle — four decades below AGC, §9-6). Squelch must not insert a faster actuator on that node. The swing-mute and V_{sq} DACs are allowed to move in ≤ 1 ms because their purpose is to *remove* the RF disturbance the heater would otherwise see; they are sequenced, not continuously adapted.

8-12 Requirements, calibration, and open items

Derived requirements

ID	Requirement	Basis
SQL-001	Squelched OMA ≤ -15 dBm (<code>Tsq_channel</code>); pre-driver+driver ≥ 20 dB electrical suppression $\rightarrow \geq 14$ dB optical vs. §8-2 max OMA -1 dBm	OCI-MSA TXO-010; Table 2-2 n.4
SQL-002	AOP constant within ± 0.5 dB of pre-squelch mission <code>Pavg</code> ; far-end demux-filter lock held; squelch shall not trigger far-end LOS (OCI-MSA RXO-006)	OCI-MSA TXO-010
SQL-003	Glitch-free three-way <code>src_sel</code> (mission / deskew / hold); serializer and TX PLL run continuously; source transitions phase-continuous so the far-end CDR does not lose lock (§6-11)	OCI-MSA deskew SM; this PMA
SQL-004	Pre-driver mute holds CM and duty-cycle balance; any CM step at the driver input settles such that launched AOP stays in the SQL-002 window	this PMA / §3-1
SQL-005	Driver swing mute while independently preserving MRM DC bias; DC operating point shall not move when RF swing is removed. Mute entry/exit ≤ 1 ms (target). Voltage-mode realization <code>TBD_analog_design</code> ; supply-collapse mute prohibited	OCI-MSA TXO-010; §3-2
SQL-006	Per-channel calibrated V_{sq} such that static transmission equals modulated-state average (nonlinear ring; not electrical mid-swing). Factory + in-service re-trim vs. TX monitor PD	OCI-MSA TXO-010
SQL-007	Junction remains in depletion at V_{sq} . Residual resonance shift inside heater linear capture range	§8-1 / §8-3
SQL-008	Heater servo remains closed-loop through entry/dwell/exit on monitor-PD average power; dither continues, amplitude bounded vs. SQL-001	OCI-MSA TXO-010
SQL-009	Servo absorbs RF-removal/restoration thermal step with zero unlock; residual detuning inside mission budget in ≤ 10 ms (target), inside the 60 ms minimum dwell	this PMA
SQL-010	Sequencing satisfies OCI-MSA Table 1-3: squelch for <code>relink_squelch_tx_duration</code> (60–75 ms) then glitch-free 160-bit training; far end detects mute within 50 ms <code>t_loselock</code>	OCI-MSA Table 1-3
SQL-011	Architecturally distinct from PMD transmit disable: disable removes AOP via ELS; squelch preserves AOP. Never implement squelch by ELS attenuation, heater park, or laser blanking	OCI-MSA DJI-003; TXO-010

ID	Requirement	Basis
SQL-012	All WDM lanes of a PMA group enter/exit together (lane-atomic). Per-lane squelch is test-only	OCI-MSA §1.1

Calibration and verification

- Per-lane V_{sq} calibration (factory and in-service re-trim) using the TX monitor photodiode; stored per lane with temperature annotation ([TBD_from_partner](#) on the monitor path and the trim engine).
- Squelched-OMA compliance at TP2 (≤ -15 dBm) and AOP-delta across entry/dwell/exit (± 0.5 dB), per lane, across temperature corners.
- Heater-lock soak: 10k+ squelch entry/exit cycles at hot and cold plate corners with zero unlock events and logged residual-detuning settling profiles.
- Partner-survival: far-end CDR and demux-filter lock monitored through squelch cycles (loss-of-modulation detect ≤ 50 ms, no LOS assertion, no far-end heater unlock) — system-level proof of SQL-002/003.
- Disable-vs-squelch discrimination: PMD transmit disable and squelch produce distinct optical signatures (AOP removed vs. AOP preserved) and cannot be cross-triggered ([SQL-011](#)).

Relationship to other Gizmo sections

Location	Relationship
§8-4 Tsqr_channel , relink_squelch_tx_duration	Optical limits this architecture implements. 60–75 ms is dwell, not enter/exit; heater lock is MRM
§3-1 serializer legal static states	src_sel = HOLD ; PLL/serializer never stop
§3-2 voltage-mode driver	Swing mute is independent of DC bias; supply-collapse mute is prohibited
§6-11 CDR signal-valid gate	This TX squelch is the far-end's invalid-signal condition; we owe phase-continuous exit and residual OMA low enough for 50 ms detect
§7-9 rule 1	Squelch does not write the heater code; disparity checker remains observe-only and is gated
§7-11 signal-invalid hold	Far-end consequence of our mute; local heater loop is the exception (it must keep running)
§7-12 / §9-4	During squelch: meas_valid low, disp_flag held; accumulators cleared on exit
§9-1 / §9-6	Entry/exit thermal step is the fully-static limit of data-dependent heating; servo absorbs it in ≤ 10 ms without unlock

Open items

- Voltage-mode realization of swing mute that does not move V_{bias} (no supply collapse, no shared swing/bias control) ([TBD_analog_design](#)).
- CDNS legal static hold code, glitch-free [src_sel](#), and CM/balance hold through mute ([TBD_from_partner](#), §3-1).
- TX monitor photodiode path, V_{sq} trim engine, temperature annotation, in-service re-trim cadence ([TBD_from_partner](#), [TBD_analog_design](#)).

- Heater-servo architecture, lock detector, linear capture range, mission vs. squelch dither profiles, integrator ramp-limit interface (TBD_from_partner; EIC digital interface TBD_analog_design).
- Confirm $t_{th,sq} \leq 10$ ms against partner τ_{th} and heater-loop bandwidth (TBD_from_sim_sweep; dwell itself stays 60–75 ms).
- Lane-group membership and lane-to-lane entry/exit skew budget for SQL-012 (TBD_from_partner).
- ELS / PMD-disable specification, and a test that disable and squelch cannot be cross-triggered (SQL-011).
- Behavioral-model TxSquelchSeq: AOP step, residual OMA including dither, heater-lock soak, and interaction with TxDisparityNrz gating (§9).
- Deskew-pattern generator placement (PMA vs. PCS) and the ≥ 285 ms training hold (TBD_from_partner on which side of the CDNS interface the 160-bit pattern lives).
- Electrical mute depth and V_{sq} must be verified at 106.25 GBd (TBD_from_sim_sweep). Protocol times (60–75 ms, 50 ms, 285 ms) are baud-independent.